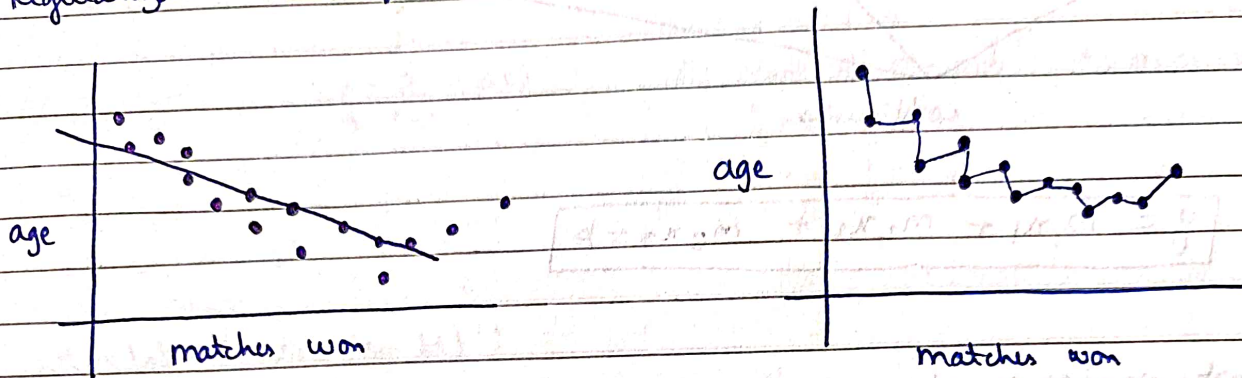


④ Regularization technique



$$\text{matches won} = \theta_0 + \theta_1 * \text{age}$$

The best fit line is not able to pass through majority of the points.

underfit



high error in case of both ~~and~~ training and testing data

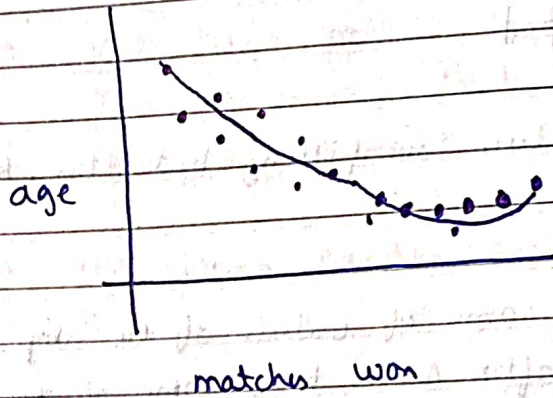
$$\text{matches won} = \theta_0 + \theta_1 * \text{age} + \theta_2 * \text{age}^2 + \theta_3 * \text{age}^3 + \theta_4 * \text{age}^4$$

The zig zag line passes through all the training points but no surety that it will pass through testing data as well.

overfit ⇒ For training data

model gives wonderful results but high error in case of testing data

balanced fit



$$\text{match won} = \theta_0 + \theta_1 * \text{age} + \theta_2 * \text{age}^2$$

$$\text{match won} = \theta_0 + \theta_1 * \text{age} + \theta_2 * \text{age}^2 + \theta_3 * \text{age}^3 + \theta_4 * \text{age}^4$$



Reduce θ_3 and θ_4 to zero

$$\text{match won} = \theta_0 + \theta_1 * \text{age} + \theta_2 * \text{age}^2$$

Ridge Regression

$$\sum_{i=1}^n (y - \hat{y})^2 + \lambda (\text{slope})^2$$

cost fn

In case of normal regression we try to reduce the value of cost fn but in case of ridge we reduce the whole thing

We are penalising (reducing the slope). Hence reduced slope reduces overfitting

Lasso regression

$$\sum_{i=1}^n (y - \hat{y})^2 + \lambda |\text{slope}|$$

This not only helps to reduce overfitting but also helps in feature elimination.

In case of ridge regression some of values of the slope gets shrunked but in case of lasso after some time some of the slopes will be equalised to zero.